The MCDM Algorithm

A Scientifically-Grounded Multi-Criteria Decision Making Engine for a Crop Rotation Decision-Aid Hub

NASA Space Apps Challenge 2026 — Crop Rotation Decision-Support Challenge

Implementation: Raspberry Pi 4 hub · NASA POWER + ISRIC SoilGrids · MODIS NDVI/EVI · FAO EcoCrop · FAO-56 · AquaCrop · AHP/SAW/TOPSIS · dynamic-programming rotation planner

Technical Report — what MCDM is, the exact mathematics, and how it is implemented in this project

1. Executive summary

Choosing a crop when several conflicting criteria matter is a **Multi-Criteria Decision Making (MCDM)** problem, and this project implements it inside a **Decision Support System (DSS)** for crop selection, following the paper the team found (Gunawan, Sitopu & Sechan, 2024). That paper's mathematics is the **weighted-sum (Simple Additive Weighting, SAW)** model: each crop is scored on several criteria; each criterion has an importance weight; the crop with the highest weighted total is recommended.

This report documents (a) what MCDM/SAW is mathematically, (b) how the paper's formulation is generalised into a rigorous, fully-specified decision pipeline — AHP weights → SAW ranking → TOPSIS verification → dynamic-programming multi-season rotation planning, and (c) how every criterion score is computed from established agronomy: the FAO EcoCrop suitability model (exact OpenCLIM formulation), FAO-56 crop water requirements, ISRIC SoilGrids soil data, and the FAO AquaCrop yield response-to-water model. All of it is driven by NASA POWER satellite data cached on a Raspberry Pi 4 for offline rural use.

The implemented engine is validated by a test suite that **reproduces the source paper's numerical example exactly** (crops A, B, C → suitability 0.76, 0.69, 0.63; winner A), alongside unit tests of every physical equation and a brute-force optimality check of the rotation planner.

Alongside the decision inputs, the hub reads a year of NASA MODIS satellite greenness — NDVI and EVI — for the chosen field. This is a **context layer, not a ranking criterion**: it helps the farmer see how the field is actually behaving, but it has no weight and never changes the crop order. Section 7 documents both indices, their exact formulas, the MOD13Q1 product, the processing pipeline, the live worked example, and the reasons for that deliberate separation.

2. MCDM inside a DSS — the method this hub implements

MCDM (also MCDA — multi-criteria decision analysis) is the discipline of making choices when several conflicting criteria matter simultaneously. The Gunawan et al. (2024) paper formalises crop selection as an optimisation:

$$\text{maximize} \quad Z = \Sigma_i \, \Sigma_j \, W_j \cdot C_{ij} \cdot X_{ij}$$

where **i** indexes crops, **j** indexes criteria, $\mathbf{W_j}$ is the importance weight of criterion j, $\mathbf{C_{ij}}$ is the suitability score of crop i on criterion j, and $\mathbf{X_{ij}} \in \{0,1\}$ selects which crop–criterion pairs count. Because exactly one crop is selected, every row uses a single X, and the model collapses to the classic **Simple Additive Weighting (SAW)** score per crop:

$$Z_i = \Sigma_j \, W_j \cdot C_{ij} \quad (\text{choose } \operatorname{argmax}_i \, Z_i)$$

The paper's worked example (weights: climate 0.3, soil fertility 0.2, market demand 0.4, water 0.1) gives $Z_A = 0.76$, $Z_B = 0.69$, $Z_C = 0.63$ → Crop A wins. Our test suite reproduces these numbers to machine precision, proving the implemented core is faithful to the paper.

What the paper leaves unspecified — and what this project therefore specifies rigorously — is: **(1)** how the weights W_j are derived from a farmer's priorities (answer: Analytic Hierarchy Process, Saaty 1980, with a consistency check), **(2)** how each C_{ij} is computed from real geophysical data (answer: EcoCrop, FAO-56, SoilGrids, AquaCrop), **(3)** whether the ranking is robust (answer: cross-check with TOPSIS), and **(4)** how to plan not just the next crop but a whole multi-season rotation (answer: dynamic programming over a documented transition-score matrix).

3. The implemented decision pipeline

Step 0 — **Data**: NASA POWER daily point series (T2M, T2M_MAX, T2M_MIN, PRECTOTCORR, WS2M, RH2M, ALLSKY_SFC_SW_DWN) and ISRIC SoilGrids soil properties (pH, sand, clay, soil organic carbon over 0–60 cm), cached locally in SQLite and refreshed every 24 hours; every feature keeps working from cache when the hub is offline. A parallel context channel fetches NASA MODIS NDVI/EVI for the same point (§7); it is displayed to the farmer but never enters the ranking.

Step 1 — **Candidate crops**: the FAO EcoCrop database (2 568 crops with temperature, precipitation, soil and cycle parameters) is bundled on the hub; the farmer shortlists crops or uses the curated default set of major food crops.

Step 2 — **Criterion scores** $c_{ij} \in [0,1]$ per crop (§4).

Step 3 — **Weights** w from AHP on the farmer's priority sliders (§5.1).

Step 4 — **Ranking**: SAW (the paper's objective), verified by TOPSIS (§5.2–5.3); both rankings are shown to the farmer.

Step 5 — **Rotation plan**: exact dynamic programming over seasons with agronomic transition scores (§6).

4. The criterion scores c_{ij}

4.1 Climate suitability — FAO EcoCrop (exact OpenCLIM formulation)

EcoCrop scores how well a crop's thermal and rainfall requirements are met. Following the OpenCLIM/ecocrop implementation (Bib. [4]), for each candidate growing window of GTIME days (GTIME stepped in 10-day increments between the crop's cycle limits GMIN..GMAX), each day t contributes an intermediate temperature score:

$$D(T) = (T - TMIN)/(TOPMN - TMIN) \;\; \text{when } TMIN < T < TOPMN$$
$$D(T) = 1 \;\; \text{when } TOPMN \le T \le TOPMX$$
$$D(T) = (TMAX - T)/(TMAX - TOPMX) \;\; \text{when } TOPMX < T < TMAX$$
$$D(T) = 0 \;\; \text{otherwise}$$

where T is the day's mean temperature and TMIN, TMAX, TOPMN, TOPMX are the crop's EcoCrop parameters. The window accumulates suitable days:

$$N_D = \Sigma_{\text{window}} \, D(T)$$

and the temperature suitability of that window is:

$$S_T = 100 \cdot [1 - (GTIME - GMIN)/(GMAX - GMIN)] \;\; \text{if } N_D \ge GMIN, \text{ else } 0$$

— i.e. the faster the crop can accumulate its minimum number of suitable days, the higher the score. Two penalties are then applied: any night below the killing temperature KTMP zeroes S_T (frost kill), and each day with maximum temperature above TMAX subtracts one point (heat stress). For perennial crops the window-mean temperature is scored instead (documented in [4]).

Precipitation suitability uses the window rainfall total PTOT:

$$S_P = 100 \cdot (PTOT - RMIN)/(½(ROPMN+ROPMX) - RMIN) \;\text{ below optimum}$$
$$S_P = 100 \cdot (RMAX - PTOT)/(RMAX - ½(ROPMN+ROPMX)) \;\text{ above optimum}$$
$$S_P = 0 \;\text{ outside [RMIN, RMAX]}$$

The combined EcoCrop score is **S = min(S_T, S_P)** — “the lowest score is likely to be the limiting factor”. The hub scans all candidate sowing windows over the cached year of NASA POWER data and reports the best window as the recommended sowing date (the OpenCLIM aggregation takes the maximum over window lengths, and a 95th-percentile when aggregating climatologies; for a single field decision the maximum is the appropriate analogue).

Rainfed vs irrigated. Under rainfed conditions the hub uses the exact OpenCLIM min(S_T, S_P): rainfall adequacy is part of climate fit. When the farmer declares an irrigation supply (mm available), the climate criterion becomes the best temperature score S_T alone — irrigation removes the rainfall limitation — and water adequacy is instead scored by the FAO-56 criterion below. This distinction is a design decision of the hub, documented because it changes interpretation in arid regions.

4.2 Soil suitability — SoilGrids vs EcoCrop soil ranges

SoilGrids 2.0 (ISRIC) provides soil pH (phh2o), sand, clay and organic carbon at standard depths; the hub averages 0–60 cm (root zone). Each property is scored with a trapezoid membership function against the crop's EcoCrop soil requirement ranges — 1 inside the optimal range (PHOPMN..PHOPMX for pH), linearly decreasing to 0 at the absolute limits (PHMIN..PHMAX); texture classes (heavy/medium/light/organic, derived from the USDA sand–clay triangle) are scored 1 if in the crop's optimal texture list, 0.5 if in its tolerated list, else 0; rooting depth is scored against the crop's DEP class. Scores are aggregated by a weighted mean (weights: pH 0.4, texture 0.3, depth 0.2, salinity 0.1 when measured EC is available):

$$S_{soil} = \Sigma_j \, w_j \, S_j, \;\; \Sigma \, w_j = 1$$

This aggregation is a design choice of the system (as the team's research notes correctly state, it is not a universal law); the per-property functions themselves follow the EcoCrop requirement ranges, and the weights are exposed and documented for calibration.

4.3 Water adequacy — FAO-56

Reference evapotranspiration is computed daily with the FAO-56 Penman–Monteith equation from NASA POWER inputs (Allen et al. 1998, Bib. [5]):

$$ET_0 = [0.408 \cdot \Delta \cdot (R_n - G) + \gamma \cdot (900/(T+273)) \cdot u_2 \cdot (e_s - e_a)] \;/\; [\Delta + \gamma \cdot (1 + 0.34 \cdot u_2)]$$

with the net radiation R_n built from the measured solar radiation R_s ($R_{ns} = 0.77 \cdot R_s$; R_{nl} via the Stefan–Boltzmann cloudiness correction), e_s from daily Tmax/Tmin, e_a from RH, and u_2 the 2 m wind speed. A Hargreaves–Samani fallback ($ET_0 = 0.0023 \cdot R_a \cdot (T_{max} - T_{min})^{0.5} \cdot (T_{mean} + 17.8)$) covers incomplete records (documented overestimation bias in arid, high-radiation conditions).

Crop evapotranspiration follows the FAO-56 stage curve (K_c initial/mid/late trapezoid, FAO-56 Table 12 values bundled for ~30 major crops):

$$ET_c = K_c(\text{stage}) \cdot ET_0$$

Effective rainfall uses the FAO “80% rule” (Dastane 1978): $P_e = 0.8 \cdot P$ for $P < 75$ mm, else $0.6 \cdot P + 15$. The seasonal net irrigation requirement is:

$$NIR = \max(0, \Sigma \, ET_c - \Sigma \, P_e)$$

In irrigated mode the water criterion is the supply fraction min(1, available/NIR); in rainfed mode it equals the EcoCrop precipitation suitability S_P (§4.1), the honest measure of rainfed adequacy. In all cases NIR is displayed as the advisory irrigation volume.

4.4 Yield response to water — simplified AquaCrop

AquaCrop's growth engine is water-driven (Steduto et al. 2009, Bib. [6]): daily transpiration translates into biomass through the normalised biomass water productivity WP*, and yield is the harvestable fraction:

$$B = WP^* \cdot \Sigma\,(Tr_i / ET_{0,i}) \qquad Y = HI \cdot B$$

WP* is conservative per functional group (C4 cereals ≈ 33.7 g m^{-2} — the AquaCrop default for maize; C3 cereals ≈ 15–20; legumes ≈ 15; the hub's bundled table gives values per group). In the simplified screening implementation, without a daily canopy–water-stress simulation, transpiration is estimated from the Kc curve as $Tr_i \approx ET_{c,i}\cdot(1 - f_e)$ with $f_e = 0.30$ (soil-evaporation share of Kc, documented). The resulting yield (t/ha) is used as a relative criterion — normalised against the best-yielding crop in the candidate set — and is explicitly labelled a screening estimate, not a calibrated AquaCrop simulation.

4.5 Rotation compatibility — pre-crop effects

Agronomic literature demonstrates that the previous crop changes the performance of the next one (pre-crop values; rotation matrices — Fenz et al. 2023 found data-driven matrices agree with literature recommendations in ~52% of pairs and within one degree in 97%; functional-group diversity raises cereal yields — Smith et al. 2023). The hub encodes a transition score T(prev → cur) ∈ [−1, 1]:

Transition	Score	Rationale
same botanical family	−1 (forbidden)	pest/disease carry-over; nutrient imbalance (family-repeat constraints are standard in crop-planning optimisation)
same crop	−0.5	monoculture penalty
legume (Fabaceae) → non-legume	+0.3	residual N credit of legumes (~30–80 kg N/ha)
deep-rooted → other	+0.1	subsoil structure / nutrient capture

The rotation criterion score used inside the MCDM matrix is $c_{rot} = ½(1 + T) \in [0,1]$.

4.6 Market demand

The farmer rates expected market demand per crop on a 0–1 slider (default 0.5, neutral). This keeps the economic criterion of the source paper (market demand, weight 0.4 in its example) in the farmer's hands, since reliable price data are not yet integrated.

5. Weights (AHP), ranking (SAW) and verification (TOPSIS)

5.1 Analytic Hierarchy Process — deriving the weights

Farmers should not have to type decimal weights — or understand criteria at all. The farmer answers five plain-language priority questions on a 1–5 scale ("not important" to "very important"): *Fits my land and weather*, *Big harvest*, *Saving water*, *Healthy soil next seasons*, and *Sells for a good price*. These map onto the six science criteria ("fits my land and weather" deliberately drives BOTH the climate and the soil criterion with the same weight — the farmer thinks of it as one question), and the 1–5 scale is stretched to the 1–9 AHP scale (p → 2p − 1, i.e. 1, 3, 5, 7, 9) so "very important vs not important" keeps its intended strength in the pairwise matrix. The hub builds the pairwise comparison matrix A with entries $a_{ij} = \text{snap}(p_i/p_j)$ to the Saaty scale {1/9, …, 1, …, 9}, and computes the **principal eigenvector**:

$$A\cdot w = \lambda_{max}\cdot w\,, \quad \Sigma\, w_j = 1$$

Consistency is measured with Saaty's index and random index RI:

$$CI = (\lambda_{max} - n)/(n - 1)\ , \quad CR = CI/RI(n)$$

with RI: n=3 → 0.58, 4 → 0.90, 5 → 1.12, 6 → 1.24, 7 → 1.32, 8 → 1.41, 9 → 1.45, 10 → 1.49. A CR ≤ 0.10 is accepted (Saaty's rule); the UI warns otherwise. The hub's typical CR ≈ 0.01 — well within tolerance.

5.2 Simple Additive Weighting — the MCDM objective

$$Z_i = \Sigma_j\ w_j \cdot c_{ij} \ \rightarrow \ \ i^* = \text{argmax}_i\ Z_i$$

This is exactly the paper's objective $Z = \Sigma\Sigma\ W_j \cdot C_{ij} \cdot X_{ij}$ under its single-crop constraint (the binary selection variables add nothing when one alternative is chosen; the argmax form is the mathematically equivalent, standard SAW statement). Because all criteria are normalised to [0,1] and $\Sigma w = 1$, $Z \in [0,1]$ is interpretable as an overall suitability.

5.3 TOPSIS — independent verification

To guard against weighting artefacts, the hub also ranks crops with TOPSIS (Hwang & Yoon 1981): vector-normalise the decision matrix ($v_{ij} = c_{ij}/\sqrt{\Sigma_i c_{ij}^2}$), weight by w_j, take the ideal a^+ and anti-ideal a^- columns, compute Euclidean distances d^+, d^- per crop and the closeness coefficient:

$$C_i = d^-_i\ /\ (d^+_i + d^-_i)$$

SAW and TOPSIS agree on the winner in practice; when they disagree the UI flags it (a genuinely borderline decision, worth the farmer's attention). All hub criteria are benefit-type (higher is better); any future cost criterion is pre-inverted.

6. Multi-season rotation planning — exact dynamic programming

Choosing the next crop is a myopic decision; a rotation must also be good over several seasons. Let Z_i be the SAW score of crop i and T(a→b) the transition score (§4.5). The hub chooses the sequence $s_1 \ldots s_K$ over K seasons (default K = 3) maximising:

$$J = \Sigma_k\ Z_{s_k} + \lambda \cdot \Sigma_{k \geq 2}\ T(s_{k-1} \rightarrow s_k)\ , \quad \lambda = 0.5$$

subject to T = −1 transitions (same botanical family) being forbidden. This is solved **exactly** by dynamic programming over a layered graph:

$$dp[1][i] = Z_i$$
$$dp[k][b] = \max_{a:\ T(a,b) > -1} \{\ dp[k-1][a] + \lambda \cdot T(a,b)\ \} + Z_b$$

with back-pointers recovering the optimal sequence; complexity $O(K \cdot n^2)$. A brute-force enumeration test confirms the DP reaches the true optimum (equivalent to a longest-path formulation; unlike greedy first-pick it never falls into locally-good, globally-poor sequences). The hub also reports a screening-level nitrogen balance over the plan: legumes credit ≈ +60 kg N/ha, non-legumes debit ≈ −45 kg N/ha per season.

7. Vegetation indices from orbit: NDVI and EVI

The hub also reads a year of satellite greenness for the selected field. This section explains the two indices — the Normalized Difference Vegetation Index (NDVI) and the Enhanced Vegetation Index (EVI) — their exact formulas, the NASA product they come from, how the hub processes them, and, just as importantly, why they are shown as context and deliberately kept out of the MCDM ranking.

7.1 NDVI — the normalized difference

Healthy leaves absorb most of the red light they receive, because chlorophyll uses it for photosynthesis, and scatter most of the near-infrared (NIR) light because leaf cell structure reflects it. The stronger the canopy, the larger the contrast between the two. NDVI turns that contrast into one

number (Rouse et al. 1974; Tucker 1979):

$$NDVI = (\rho_{NIR} - \rho_{red}) / (\rho_{NIR} + \rho_{red})$$

where ρ_{NIR} and ρ_{red} are surface reflectances in the near-infrared (MODIS band 2, 841–876 nm) and red (band 1, 620–670 nm) wavelengths. The ratio is normalized to [−1, 1]. Negative values indicate water, snow or cloud; values near zero indicate bare soil, rock, dry stubble or built-up land; positive values rise with the amount of living green leaf area. For a field, the seasonal shape — green-up, peak, senescence — is usually more informative than any single number.

NDVI	What the value usually indicates (indicative only)
below 0	water, snow, ice, or a cloud-contaminated pixel
0.0 – 0.15	bare soil, rock, dry stubble, roads and buildings
0.15 – 0.30	sparse vegetation, newly emerged or senescing crops
0.30 – 0.60	developing to full crop canopy
0.60 – 0.90	dense, healthy canopy (irrigated crops, forest)

NDVI is simple, long-running and comparable across sensors, which is why it became the continuity index for global vegetation monitoring. It has three well-known weaknesses: it **saturates** over dense canopies, because red reflectance is already fully absorbed and extra leaf area barely changes the ratio; the **soil background** biases it at low cover; and **aerosols and thin cloud** affect the red and NIR bands differently. EVI was designed to address these.

7.2 EVI — the enhanced vegetation index

EVI keeps the same red/NIR contrast but adds a blue band and three correction terms (Huete et al. 2002):

$$EVI = G \cdot (\rho_{NIR} - \rho_{red}) / (\rho_{NIR} + C_1 \cdot \rho_{red} - C_2 \cdot \rho_{blue} + L)$$

with the coefficients adopted for MODIS: **G = 2.5** (gain, which scales the index), **C_1 = 6** and **C_2 = 7.5** (aerosol-resistance coefficients), and **L = 1** (canopy-background adjustment). The blue band (MODIS band 3, 459–479 nm) is used because aerosol scattering is stronger in the blue than in the red: the C_1/C_2 terms estimate and remove the part of the red signal caused by haze rather than by the canopy. L compensates for the non-linear transfer of red and NIR radiation through a canopy and for the brightness of the soil below it. G brings the result into a familiar range.

EVI is generally lower than NDVI for the same canopy, but it keeps responding where NDVI has saturated, and it is much less sensitive to a bright soil background or a hazy atmosphere. Over very bright targets — snow, ice, thick cloud — the three-band EVI becomes unstable, so the MODIS algorithm substitutes a two-band EVI (without the blue term) for those pixels; the hub consumes the EVI layer as the product delivers it. Neither index is “more correct” in general: they are two views of the same canopy, which is why the hub shows both.

The Cairo example in §7.4 makes the difference concrete. There NDVI is 0.17 while EVI is only 0.09. The field sits on the desert fringe, where bright soil and airborne dust inflate the simple red/NIR ratio; EVI's soil and aerosol corrections remove much of that signal. A global NDVI threshold would call this field “sparse vegetation”; relative to its own year it is in fact at its greenest. That is exactly why the hub compares each field with itself, not with a world map.

7.3 The data product and the hub pipeline

Both indices come from one product, so they describe the same observations: **MODIS/Terra MOD13Q1, Collection 6.1**, the standard NASA 250 m, 16-day vegetation-index composite (Didan 2021; MODIS VI User Guide 2015). The 16-day maximum-value composite selects, for each pixel, the best observation from all acquisitions in the period using low cloud, low view angle and the highest index value — a practical way to suppress cloud contamination.

Property	MOD13Q1 value used by the hub
Product	MODIS/Terra Vegetation Indices, 16-day L3 Global 250 m SIN Grid, v061
Spectral bands	red 620–670 nm (band 1), NIR 841–876 nm (band 2), blue 459–479 nm (band 3, EVI only)
Spatial resolution	250 m — one pixel covers ~6.25 ha at the equator
Temporal composite	16 days; ~22–23 complete composites per year
Layers requested	250m_16_days_NDVI and 250m_16_days_EVI
Storage encoding	int16, scale factor 0.0001, fill −3000, valid range −2000…10000
Access	ORNL DAAC MODIS/VIIRS Subsetting Tool REST API, no token required
Coverage	global land, 2000–present
Hub cadence	12-month series per field, cached 7 days

The pipeline is deliberately small and reproducible:

Step	What the hub does
1. Request	one pixel at the field's latitude/longitude (kmAboveBelow = 0, kmLeftRight = 0), one year back from today
2. Window	the API returns at most 10 composites per call, so the year is split into three ~150-day windows for each band (six calls, run in parallel; ~10 s uncached)
3. Decode	raw int16 × 0.0001; discard the fill value −3000 and anything outside the valid range −2000…10000
4. Pair	keep a date only when NDVI and EVI are both valid — the two indices must come from the same composite
5. Cache	SQLite dataset mod13q1_veg: 7-day time-to-live, an error payload is retried after 1 h, and the recommendation keeps working offline without vegetation data
6. Analyse	latest values, percentile within the field's own year, change since the previous composite, seasonal peak and trough with dates, number of composites
7. Present	the “From orbit” card and seasonal chart, always with the composite date and the product attribution

The values are exposed by **GET /api/vegetation?lat=…&lon=…**. The response carries the series, the latest values, the statistics, the trend, the source string and the count; when the service is offline the endpoint returns an error object and the card degrades to a plain message — the crop advice is never blocked by it.

7.4 Reading the numbers

A single index value has no universal meaning: 0.4 can be a good year in a temperate maize field and a very good year on a dry rangeland. The hub therefore reports **relative** measures computed from the field's own 12 months:

Output	Definition used in the hub
Latest NDVI / EVI	the most recent complete composite, with its date
Percentile	share of the field's own composites in the past year with a lower value (0–100); “greenest reading in the past year” when ≥ 99
Trend	difference from the previous composite; a change larger than ±0.02 is called greener or drier, otherwise steady (a deliberately conservative threshold for a 250 m mixed pixel)
Peak / trough	highest and lowest index value in the year, with the dates
Count	number of valid complete composites behind the summary

Live worked example — the default field in the app (30.0444 N, 31.2357 E, Cairo) on 13 November 2026: 22 complete composites. The latest, for 16 October 2026, is NDVI 0.17 and EVI 0.09. NDVI ranges from 0.07 (2 February 2026) to 0.17, with the peak on 16 October; EVI peaks at 0.09 on 9 May. The latest NDVI is the highest of the 22 composites (100th percentile), and the change since the previous composite is +0.038, so the card reads “the field is greening since the previous 16-day reading”. The absolute values are low — typical of a dry, sparsely vegetated 250 m pixel on the desert

fringe — but the seasonal signal is real and is read against this field's own baseline. The same run also illustrates the NDVI/EVI difference: the bright soil background lifts NDVI more than EVI.

7.5 How the indices are used in this project — and what they are not used for

Used for. The pair is a field-condition panel: it shows how green the field is now and how that greenness moved through the year, so the farmer can notice drought stress, delayed green-up, an early senescence, or a field that is not doing what they expected. It is also a transparency device: the composite date, the product and the source are printed with the chart, so the farmer can see exactly what the satellite saw and when.

Not used for. NDVI and EVI are **not criteria** in the MCDM model. They do not appear in c_{ij}, they carry no weight, and they do not enter the SAW score, the TOPSIS check or the rotation dynamic program. This is a deliberate design decision, documented so that the ranking can be audited: every number that influences the ranking has a weight shown in §5, and the vegetation indices have none. There are four reasons:

(1) Double counting. The climate and water criteria already encode the site's temperature and rainfall from NASA POWER. Greenness is largely a response to those same drivers; adding it as a criterion would count one signal twice and silently inflate its effective weight.

(2) Target mismatch. NDVI and EVI measure the vegetation that is on the ground now, not the suitability of a candidate crop. A fallow field with NDVI 0.1 can be an excellent choice for wheat; a green patch of weeds says nothing about whether a crop fits. Suitability is a forward-looking question about requirements, and it is answered by EcoCrop, FAO-56 and SoilGrids.

(3) Temporal mismatch. The recommendation is for the next season; the latest composite describes the previous 16 days. Using a past observation as a future suitability score would confuse state with forecast.

(4) Spatial mismatch. A 250 m pixel mixes the field with tracks, neighbouring plots, bare patches and buildings. It cannot support a field-level suitability claim. It can, however, support a conversation — which is precisely how the card presents it.

Keeping the indices outside the ranking also protects the project's plain-language promise: the farmer never has to understand a remote-sensing index to understand why a crop was recommended. The greenness panel answers a different question — “how is my field doing?” — and says so on its face.

7.6 Limitations specific to the vegetation panel

1. **Mixed pixels.** At 250 m, one value covers ~6.25 ha. A small field shares its pixel with roads, neighbours and bare ground; the panel is a landscape-scale context, not a per-field measurement.

2. **Composite timing.** A 16-day maximum-value composite describes the best observation in that window, not a specific day. Events shorter than the window — a single heat spike, a flash flood — can be smoothed away.

3. **Quality screening.** The hub keeps values inside the product's valid range and requires both NDVI and EVI on the same date, but it does not yet decode the detailed pixel-reliability/VI-quality layers. Residual thin cloud or aerosol can therefore survive in a composite; the trend note is worded conservatively for that reason.

4. **Index behaviour.** NDVI saturates in dense canopies; EVI is noisier over bright targets (snow, ice, thick cloud), where MODIS switches to a two-band EVI. The hub displays the product's EVI layer as delivered and does not attempt its own atmospheric correction.

5. **Green is not crop.** Neither index distinguishes a crop from weeds, cover crops or volunteer vegetation. A green field is not necessarily a well-managed field.

6. **No absolute calibration.** The indices are not biomass or yield. They are compared within the field's own year and shown as relative context, not converted into kilograms or tonnes.

7. **One-year window.** Twelve months is enough to characterise the seasonal cycle but not to establish multi-year degradation or a long-term trend; the hub does not claim one.

8. The hub system

Component	Implementation	Notes
Hardware target	Raspberry Pi 4	backend: Python stdlib + numpy (+ requests for data fetches)
Web interface	Farmer-friendly single-page app, port 8000	plain language throughout; no math exposed; four numbered steps; 0–5 scorecard bars instead of raw scores; expandable "What's this?" explanations state how every factor is measured (NASA POWER weather, SoilGrids soil map, FAO EcoCrop requirements, FAO-56 water use, AquaCrop harvest); market demand is four labelled choices (Hard to sell / Average / Good / Sells fast), not a numeric slider; water supply is "Rain only" or a volume choice with the helper "1 mm = 1 litre per square metre"; offline MAP PICKER (tap/pinch) shows DAILY NASA satellite imagery via GIBS/Worldview with a parameter filter (true colour, SMAP soil moisture, MODIS land temperature, VIIRS NDVI greenery, IMERG rainfall — soil pH comes from SoilGrids, not daily satellites); NASA/FAO/ISRIC attribution and provenance (date range, soil pH) shown with every result; no emojis; SVG leaf favicon
Deployment	systemd service + venv (INSTALLED and verified)	scripts/install.sh creates the venv (numpy), installs systemd/lcrda.service (auto-restart, verified by SIGKILL test), and a daemon thread re-pulls NASA POWER for every cached location every 24 h
Climate data	NASA POWER daily point API (7 variables)	cached in SQLite; 24-hour refresh (background thread); one request per 0.5° cell (NASA fair-use)
Map imagery	NASA GIBS WMTS (MODIS true colour, SMAP soil moisture, MODIS LST, VIIRS NDVI, IMERG rain)	daily imagery through the hub proxy /api/maptile, disk-cached per day (offline = newest cached day, Natural Earth outlines as base layer); tiles older than 8 days auto-purged; layer dates auto-resolved (lagging products probed at their likely date)
Vegetation context	NASA MODIS MOD13Q1 v061 (NDVI + EVI, 250 m, 16-day) via the ORNL DAAC subsetting API	one pixel at the field, 12-month series, cached 7 days; shown as the “From orbit” card with a seasonal chart and composite date; explicitly NOT an MCDM criterion — see §7
Soil data	ISRIC SoilGrids 2.0 REST (pH, sand, clay, SOC, 0–60 cm)	cached 30 days; handles NULL cells (rivers, dense cities) by sampling the nearest valid cells within ~30 km; error payloads retry after 1 h, never live a full TTL
Crop parameters	FAO EcoCrop database, 2 568 crops	bundled CSV (FAO EcoCrop / OpenCLIM, OGL v3)
Decision engine	AHP → SAW → TOPSIS + DP rotation	fully deterministic, reproducible; when SAW and TOPSIS disagree the UI says the choice is a close call and names the runner-up
Offline behaviour	all features run from cache	designed for farmers without internet

API surface: GET /api/health, GET /api/crops?q=…, GET /api/vegetation ?lat=…&lon=… (12 months of MODIS NDVI/EVI context, never part of the ranking), POST /api/recommend (lat, lon, shortlist, farmer priorities {land, harvest, water, soilnext, market} on 1–5 — raw 1–9 criterion keys also accepted, previous crop, water available in mm (null = rainfed), market scores 0.2–1.0; the response echoes waterMode/waterAvailableMM and includes closeCall when the two methods disagree), POST

/api/refresh (fair-use guarded: refetches only when the cache is stale).

8.1 Validation

Test	Expected	Result
Paper numerical example (SAW)	Z_A=0.76, Z_B=0.69, Z_C=0.63, winner A	PASS (machine precision)
EcoCrop daily/window equations	piecewise scores, frost zeroing	PASS
FAO-56 ET_0 / Kc / NIR	plausible ET_0, Kc trapezoid, Pe rule	PASS
AquaCrop B = WP*·Σ(Tr/ET_0), Y = HI·B	exact algebra	PASS
AHP consistency diagnostics	CR ≤ 0.1 detection	PASS
Rotation DP vs brute force	identical optimum	PASS
MODIS NDVI/EVI client	scale 0.0001, fill −3000 / valid −2000…10000, same-date NDVI+EVI pairing	PASS
End-to-end live run (Egypt, 29.2 N 31.0 E)	winter crops for Nov sowing; AHP CR 0.0092; SAW=TOPSIS winner	PASS

43/43 tests pass (mcdm-hub/tests/test_mcdm.py), including the NDVI/EVI decoding, pairing and summary-analysis tests. The end-to-end run used live NASA POWER data (366 days cached, ET_0 mean 6.79 mm/day) and live SoilGrids data (pH 8.0, clay 24.8 %, sand 47.5 %, SOC 4.8 %).

9. Known limitations (documented, not hidden)

1. **Screening-level yield**: the AquaCrop implementation is the conceptual equation without the full daily soil-water–canopy simulation; treat yields as relative indicators. Calibrate with local data before claiming quantitative accuracy (the team's own Resources.md demands this).

2. **Simplified seasonal water balance**: NIR uses effective rainfall, not a daily soil-water-bucket model; adequate for ranking, not for irrigation scheduling.

3. **SoilGrids is modelled data** (~1 km): the hub accepts measured pH (and could accept EC) to override.

4. **Weights and transition scores** are design choices documented for calibration, not universal laws — consistent with the MCDM literature's advice to elicit weights from stakeholders.

5. **EcoCrop rainfall suitability is rainfed-oriented**: hence the explicit rainfed/irrigated mode (§4.1).

6. **Market criterion** is farmer-entered until a price data source is integrated.

7. **24-hour cache** is a fairness/simplicity trade-off; POWER updates daily, so no information is lost.

8. **Vegetation indices are context, not criteria.** NDVI/EVI are 250 m mixed-pixel observations of current land cover; §7.5 explains why they carry no weight in the ranking and §7.6 lists their own limitations (composite timing, QA, saturation, one-year window).

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Appendix A — repository map: mcdm-hub/run.py (entry) · backend/engine/ (mcdm, ecocrop, fao56, soil, aquacrop, rotation) · backend/data/ (power, soilgrids, crops + EcoCrop_DB.csv, vegetation) · backend/storage.py (SQLite cache) · backend/app.py (pipeline) · backend/server.py (HTTP) · frontend/ (UI) · tests/test_mcdm.py (43 tests) · agent.md (full research & work log).